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Transcranial direct current stimulation enhances mood but not performance in elite athletes

Brief title: tDCS and swimming performance

Submission type: Original investigation

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Abstract

Purpose: This study aimed to determine if transcranial direct current stimulation (tDCS) could be effective for the enhancement of swimming performance or mood state in elite athletes. **Methods:** Eight male elite triathletes (age, 20 ± 2 years; maximal oxygen uptake, 71 ± 4 $ml \cdot kg^{-1} \cdot min^{-1}$) participated in this crossover, counterbalanced, sham-controlled, double-blind study. Participants received either actual (20 minutes of anodal stimulation of the motor cortex at 2 mA) or sham tDCS, and performed an 800-meter swimming test in which rating of perceived exertion and blood lactate response were measured. Mood state (Brunel Mood Scale, BRUMS) was assessed before and after each tDCS session, and after the swimming test. Heart rate variability and central nervous system readiness were assessed before and after each tDCS sessions. The chances of finding differences between conditions were determined using magnitude-based inferences. **Results:** A significant and very likely higher BRUMS-determined vigor self-perception was found with actual tDCS after the stimulation session (-0.1 ± 1.2 and 2.0 ± 2.3 for sham and actual tDCS, respectively; p=0.018, effect size (ES)=1.14) and after exercise (-4.1 ± 2.9 and -0.9 ± 3.6, respectively; p=0.022, ES=0.98). However, likely trivial and non-significant (p>0.05) differences were found between conditions in performance (599 ± 38 and 596 ± 39 seconds, respectively). Unclear and non-significant differences were observed between conditions for the rest of study endpoints. **Conclusions:** tDCS elicited a marked increase in vigor self-perception that was also maintained after exercise, but failed to improve swimming performance in elite triathletes.

Key words: endurance performance, fatigue, mood, brain stimulation, central fatigue.

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Introduction

Exertional fatigue is a physical and perceptual reaction that results in an increased effort to exert a given force and a decline in the ability to exert this force.[1,2] Although peripheral mechanisms contribute to fatigue development during exercise, reductions in neural drive (*i.e.*, supraspinal fatigue) with subsequent decrease in spinal motoneuron excitability also play a pivotal role.[1,2] In this context, increasing motor cortex excitability by means of non-invasive brain stimulation has been suggested to enhance physical and mental performance.[3] Together with transcranial magnetic stimulation, a main non-invasive brain stimulation technique is transcranial direct current stimulation (tDCS),[3] which consists of delivering a weak continuous electrical current to a targeted brain area using surface electrodes attached to the scalp. When applied over the motor cortex, tDCS has proven to increase corticospinal excitability [4] and sensory and pain thresholds in healthy individuals.[5] As both decreases in pain tolerance[6] and corticospinal excitability[1,2] are related to the appearance of fatigue during exercise, tDCS might potentially enhance performance.

Although tDCS is gaining popularity for its potential ergogenic effect on athletic performance,[7] there is yet not enough scientific evidence to support its effectiveness, especially in the real-world sport context. Most studies have analyzed the effects of this technique on single-joint isometric exercises (*e.g.*, knee extension, elbow flexion),[7] and only in recent years have its effects on whole body dynamic exercise (*i.e.*, cycling) been tested.[7] Some benefits of tDCS have been reported, such as improvements in performance itself (*e.g.*, power output or time to exhaustion) or decreases in the perceived effort for a given external load.[8–10] However, the effectiveness of tDCS for the enhancement of performance in maximal incremental or time to exhaustion tests remains unclear,[8–12] and to our knowledge only one study has analyzed its effects during self-paced exercise.[12] Moreover, most research has been conducted in active or untrained subjects[8,10–12] and only one study to date has assessed trained athletes (cyclists).[9]

Therefore, there is little or no evidence on the effectiveness of tDCS for enhancing endurance performance in elite athletes or in self-paced exercises. The main goal of our study was to analyze if tDCS could improve swimming performance (primary endpoint) in elite athletes. We also studied tDCS

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effects on blood lactate concentration, effort and mood self-perception, cardiac autonomic modulation, and central nervous system (CNS) readiness (secondary endpoints).

Methods

Participants

Eight male elite triathletes volunteered to participate in this study (age [mean ± SD], 20 ± 2 years; body mass index, 21 ± 1 kg·m^{-2}; body fat, 11 ± 1%; maximal oxygen uptake, 71 ± 4 ml·kg^{-1}·min^{-1}). All of them were healthy, highly trained (~26–27 hours·week^{-1}) and competed at international level. Owing to the gender differences in the current received during tDCS, with women receiving less current at targeted parietal cortex than men when stimulated at identical current levels,[13] only men were included in this study. The study was performed during the winter season, before the start of the competition period. Participants refrained from taking ergogenic aids or stimulants (*e.g.*, caffeine) as well as from doing intense exercise during the 48 hours prior to each testing session. They were instructed to maintain their normal dietary patterns during the study. Participants signed an informed consent form after the procedures had been explained in detail. The study was approved by the Local Ethics Committee (*University of Alcalá*, Madrid, Spain).

Design

A crossover, counterbalanced, sham-controlled, double-blind design was used. A schematic representation of the experimental protocol is shown in **Figure 1a**. Participants attended the testing facilities on two different occasions separated by one week. During each visit they received either actual or sham tDCS. Each participant was tested at approximately the same time of the day (always in the morning).

Procedures

The tDCS session consisted of 20 minutes of anodal tDCS (BrainSTIM, E.M.S, Italy) at a constant current of 2 mA with a 30-second ramp at on- and offset using two 5 × 5 cm sponge electrodes soaked in 0.9% sodium chloride. The anode was placed on the C3 location in the International 10:20 System and the cathode was placed on the contralateral supraorbital Fp2 location (**Figure 1b**). The research staff had undertaken validated training in landmarking the tDCS stimulator pads before the

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start of the protocol, which has previously proven to increase cortical excitability.[14] During the sham tDCS condition all procedures were performed as above mentioned, but the stimulation lasted only 15 seconds with the same 30-second ramps. This sham tDCS method does not cause neuro-modulatory effects but has been reported to elicit sensations that are similar to those caused by actual tDCS stimulation.[15] The fact that the tDCS equipment we used only works when appropriate impedance is achieved by generating an acoustic signal if impedance is impaired during the stimulation session, ensured that stimulation was correct during throughout the protocol. Participants and researchers in charge of tDCS administration, endpoint assessment or statistical analysis were blinded to the nature of the tDCS intervention (actual or sham). After each stimulation session participants were asked whether they thought they had received actual or sham tDCS. The occurrence and severity of tDCS-related adverse effects was also assessed.[16]

After the tDCS session participants performed an 800-meter freestyle swimming test which was preceded by a standardized warm-up (*i.e.*, 400 meters of light swimming freestyle, followed by four repetitions of 25 meters all out interspersed with one-minute rests, and 200 meters of light swimming). The start of the swimming test was marked with an acoustic signal, with participants inside the water. They were encouraged to complete the test as fast as possible, but no specific cues regarding pacing were provided. Two experienced coaches recorded swimming times each 100 meters with a stopwatch (3x100, Finis, USA) and the mean of both measurements was used for analyses (difference between raters for all measurements=0.08 ± 0.57 seconds; intra-class correlation coefficient=0.99). Speed was analyzed both as raw (meters·second^{-1}) and normalized data (%). Normalized speed (%) was calculated by relativizing raw speed to the average speed of the test. It has been proposed that analyzing the difference between current and overall mean velocity reflects the pacing pattern better than overall performance.[17] The fatigue index was calculated as the ratio between the raw speed of the first and second halves of the test. The rating of perceived exertion (RPE) after the swimming performance test was determined on a 1 to 10 scale (Borg CR-10 scale).

Fingertip capillary blood samples (0.7 µl) were taken immediately before, and 3, 5 and 9 minutes after the swimming test, respectively. Lactate concentration was measured with a portable

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lactate analyzer (Lactate Plus, Nova Biomedical, USA). The peak blood lactate concentration observed after each test was used for statistical analyses.

Mood state was assessed with the Brunel Mood Scale (BRUMS)[18] immediately before and after the tDCS session as well as 5 minutes after the swimming performance test. This questionnaire contains 24 simple mood descriptors divided into six domains (confusion, anger, depression, fatigue, tension, and vigor) and participants indicate whether they are experiencing such feelings on a five-point Likert scale (0 = not at all, 1 = a little, 2 = moderately, 3 = quite a bit, 4 = extremely). Data were normalized for the analysis by subtracting the baseline score to those obtained after the tDCS session and the swimming performance test, respectively.[8]

Heart rate variability (HRV) was calculated from heart rate measurements taken at baseline and immediately after the tDCS session. During measurements, athletes were lying supine and relaxed for 10 minutes, and all R–R intervals during the last 5 minutes were analyzed. The natural logarithm of the root-mean-square differences of successive heart beat intervals (LnRMSSD) was calculated with a specific software (Kubios HRV software, Finland). The LnRMSSD has been described as the most reliable and practically applicable measure for the monitoring of athletes.[19]

CNS readiness was analyzed at baseline and after the tDCS session via determination of direct current (DC) potential using a novel device (Omegawave Ltd, Espoo, Finland). The DC potential, defined as brain bi-potentials within a frequency range (0–0.5Hz) lower than electroencephalogram frequencies, has been reported as a marker of brain status, being related to functions such as task demand and performance, alertness or attention.[20] Two electrodes were connected to the sensor and placed on the athlete's forehead and on the base of the dominant hand's thumb. As with HRV, the measurement was performed with the athlete relaxed and lying supine. A proprietary algorithm then generated a CNS readiness score that ranged between 1 and 7 arbitrary units (AU), with higher scores reflecting a better performance.[21]

Statistical analysis

The 800-meter swimming test performed in our study was routinely performed as a regular performance assessment (at least once a month) by the triathletes enrolled in our study, who were thus

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familiarized with the test. Based on two tests performed the month before the experiments (594 ± 36 and 597 ± 38 seconds for the first and second trial, respectively; retest correlation coefficient [r]=0.975), *a priori* calculations were performed using a spreadsheet[22] to determine the minimal sample size needed to obtain clinically meaningful differences $(\text{i.e.}, 0.20 \times between\ subject\ SD)$. Considering the within-subject SD $(between\ subject\ SD \times \sqrt{(1-r)})$ and the between-subject SD, the minimal required sample size was 7 subjects for a chance of type I and type II error ≤0.5% and ≤25%, respectively.

Data are presented as mean ± SD. The normal distribution (Shapiro-Wilk test) and homoscedasticity (Levene´s test) of the data was checked before any statistical treatment. Differences in proportions were evaluated using the Fisher's exact test. Differences between conditions (actual *vs.* sham tDCS) in swimming performance time, peak blood lactate and RPE were analyzed using paired Student's t-tests. A two-way (condition x time) repeated-measures ANOVA was used to assess differences in swimming pacing (eight time points), HRV (two time points), CNS readiness (two time points) and BRUMS scores (three time points). Greenhouse–Geisser correction was applied when Mauchly's Test of Sphericity was violated, and *post hoc* analysis (Bonferroni test) was conducted when a significant interaction (condition × time) effect was found. The effect size (ES) of the significant differences was calculated as partial eta-squared (η_p^2) for condition x time interactions or as standardized mean differences (Hedges' g) for specific time points, and the obtained ES values were considered small (ηp^2=0.01, g=0.20) moderate (ηp^2=0.06, g=0.60) or large (ηp^2=0.14, g=1.20).[23,24] Analyses were performed using a statistical package (SPSS version 23.0, IBM, NY) with the significance level set at 0.05. The chances of finding differences between conditions were assessed with specific spreadsheets[25] using magnitude-based inference and considered as follows: <1%, almost certainly not; 1–5%, very unlikely; 5–25%, unlikely; 25–75%, possible; 75–95%, likely; 95–99%, very likely; and >99%, almost certain.[23] If the chances of having better and poorer results were both >5%, the difference was considered unclear.

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Results

Blindness and adverse effects

No significant differences (p>0.05) were observed between conditions (actual *vs.* sham tDCS) in the percentage of participants who thought they had received the sham intervention (50% and 25% for sham and actual tDCS, respectively), the real intervention (25% and 50%) or that did not know what to answer (25% and 25%).

No significant differences (p>0.05) were observed between conditions in the prevalence or severity of adverse effects (**Table 1**). A light-to-severe tingling sensation was the most commonly reported adverse effect, being present in all the participants with both stimulation conditions. Some cases of light sleepiness, skin redness and headache were also reported.

Effects of tDCS

Scores on the BRUMS scale are shown in **Table 2**. A significant time and condition x time interaction effect with a large ES (η_p^2=0.47) was observed for vigor. *Post hoc* analysis revealed a very likely and significantly higher vigor with actual tDCS compared to sham tDCS after both stimulation (*p*=0.018, g=1.14) and exercise (*p*=0.022, g=0.98). A significant time effect but no significant condition x time interaction was observed for fatigue. No significant time or condition x time interaction effect was observed for anger, confusion, depression or tension.

Likely trivial and non-significant (*p*=0.346) differences were found between conditions for overall swimming performance (599 ± 38 and 596 ± 39 s for sham and actual tDCS, respectively). A significant time effect (p<0.001) but no significant (0>0.05) condition x time interaction effect was observed for raw (time effect: $F_{(1.902,\ 13.313)}$=41.920, *p*<0.001; condition x time effect: $F_{(7,49)}$=1.996, *p*=0.075; **Figure 2a**) and normalized speed (time effect: $F_{(1.844,\ 12.909)}$=38.457, *p*<0.001; condition x time effect: $F_{(7,49)}$=2.084, *p*=0.063; **Figure 2b**). A likely and almost significantly lower fatigue index (*p*=0.058) with a moderate ES (g=0.63) was observed with actual tDCS (1.03 ± 0.02 and 1.02 ± 0.01 for sham and actual tDCS, respectively).

Unclear and non-significant differences were found for baseline (1.2 ± 0.4 and 1.3 ± 0.6 $mmol \cdot l^{-1}$ for sham and actual tDCS, respectively; *p*=0.53) and peak capillary blood lactate levels (9.9 ± 3.1 and

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9.4 ± 2.4 mmol·l^{-1}, respectively; *p*=0.48). Unclear and non-significant differences were also found between conditions for RPE (9 ± 2 and 9 ± 1 for sham and actual tDCS, respectively; *p*=0.47).

The HRV of one of the participants could not be analyzed due to excessive artifacts. The LnRMSSD for sham and actual tDCS was 3.8 ± 0.8 and 3.9 ± 1.1 ms, respectively, at baseline; and 3.8 ± 1.1 and 3.9 ± 0.8 ms, respectively, after the tDCS session. Unclear and non-significant (time effect: $F_{(1,6)}$=0.273, *p*=0.620; condition x time effect: $F_{(1,6)}$=0.079, *p*=0.788) differences were observed between conditions. The CNS readiness for sham and actual tDCS was 5.4 ± 1.3 and 5.6 ± 1.3 AU, respectively, at baseline; and 5.4 ± 2.1 and 6.1 ± 0.8 AU, respectively, after the tDCS session. Unclear and non-significant differences (time effect: $F_{(1,7)}$=0.467, *p*=0.516; condition x time effect: $F_{(1,7)}$=0.212, p=0.659) were observed between conditions.

Discussion

tDCS has proven to increase corticospinal excitability[4] as well as to decrease pain perception,[5] with both factors being related to the appearance of exercise-related fatigue.[1,2] In this context, we hypothesized that tDCS could enhance performance during a self-paced exercise such as swimming. Yet, our findings refute our original hypothesis.

Contrary to ours and other reports,[11,12] a beneficial effect of tDCS on whole-body exercise has been observed in some studies, such as an improved performance (*e.g.*, power output or time to exhaustion) or a decreased perceived effort for a given external load.[8–10] This disparity of results could be caused, at least partly, by the large individual variability in the effects of tDCS on corticospinal excitability, possibly as a result of individual anatomical differences.[26] Moreover, it has been suggested that the controversy on the effectiveness of tDCS for the enhancement of performance may also be due to the different electrode montages (*i.e.*, cephalic *vs*. extra-cephalic montage) adopted in the scientific literature.[7] Thus, placing the cathode over the shoulder might be more effective than its placement over the dorsolateral prefrontal cortex, as the latter may impair the effect of the anode over the motor cortex.[27] Notably, as confirmed by recent evidence,[10] an extra-cephalic montage –and not a cephalic one such as that used here– seems effective for the improvement endurance performance.

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It has been reported that exercising (*i.e.*, leg extension) to fatigue during ~ 10 minutes (a duration similar to that of the swimming test performed in the present study) results in a decreased output of spinal motoneurons,[28] which could be potentially attenuated by tDCS. However, the contribution of central and peripheral mechanisms to fatigue during self-paced exercise is duration-dependent. Thomas et al.[29] observed a greater and lower peripheral and central fatigue, respectively, with short-duration cycling exercise (~6 minutes) than with more prolonged bouts (>30 minutes). Thus, we hypothesize that tDCS could provide greater benefits on those exercise tasks where central mechanisms play a major role on fatigue development, that is, on activities of longer duration.

Notwithstanding, in the present study tDCS resulted in a higher vigor (which corresponds to a higher score in the BRUMS items *lively*, *energetic*, *active* and *alert*), being this effect maintained after exercise. Subjective measures like the BRUMS have proven superior sensitivity and consistency for monitoring athletes' well-being than objective measures such as blood markers,[30] and a relationship between mood state (particularly vigor self-perception) and athletic performance has been observed.[31,32] Brandt et al.[32] stated that new ways to maintain high levels of vigor in athletes should be explored, as lower levels of this mood status indicator are associated with a decreased chance of sporting success. To our knowledge, ours is the first study to report an improved mood state in athletes after tDCS. These results could be of great interest for athletes, as tDCS might potentially serve to improve vigor sensation before a competition. Further research is needed, however, to confirm the relationship between mood and performance.

Previous research in non-athletic (clinical) populations supports the beneficial effect of tDCS for the improvement of mood state.[33,34] Kheder et al.[33] observed a mood improvement in patients with fibromyalgia related to increased serum beta-endorphins levels after anodal tDCS applied over the motor cortex (C3 in the 10–20 EEG system, like in our study; **Figure 1b**). tDCS not only alters the functional connectivity of the region under the anode (in our case, the motor cortex), but also that of other cortical and deep brain regions such as the dorsolateral prefrontal,[35,36] especially with the conventional methodology using two large sponge pads such as the ones used here (25 cm^2). Thus, the spread of the electrical current from the left motor cortex (C3 in the 10–20 EEG system, see **Figure 1b**)

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to the left dorsolateral prefrontal cortex (F3 in the 10–20 EEG system, see **Figure 1b**), which plays a key role in mood and emotional processing,[37] could explain the observed improved mood after tDCS.

On the other hand, no changes were observed on CNS readiness or HRV. It has been suggested that tDCS might influence cardiac autonomic regulation due to the role of the insular cortex on autonomic nervous activity.[38] However, controversy exists regarding the effects of tDCS on HRV, with studies finding a shift towards vagal modulation,[39] sympathetic dominance,[40] or other authors reporting no changes.[38] Moreover, as tDCS has proven to increase brain excitability[4], we hypothesized that it could influence the DC-potential, a marker of brain function that has been related to task performance, alertness or attention.[20] However, no changes were observed in CNS readiness (which is calculated from DC-potential), at least with the user-friendly device employed in this study.

Practical applications

Anodal tDCS applied unilaterally over the primary motor cortex is safe and seems effective for the improvement of vigor self-perception in elite athletes, which might be explained by an increased excitability of the motor cortex but also by the involuntary stimulation of other adjacent brain areas such as the left dorsolateral prefrontal cortex. However, this strategy provided no benefits on swimming performance.

Our study has some limitations. We did not recruit many participants, which we believe is justifiable given the difficulty of conducting research with high-performance athletes such as those assessed here (*i.e.*, elite triathletes competing at international level). It is also noteworthy that our sample size was big enough to obtain meaningful results. Another limitation was the lack of measurements of cortical excitability or brain activity, which could have provided a deeper insight on the actual physiological effects of the tDCS protocol we used. Indeed, it has been reported that slight changes in this tDCS protocol (*i.e.*, stimulating with 2 mA instead of with 1 mA) can result in opposite effects on cortical excitability.[14] In turn, major strengths and novelties of our approach was the fact that we assessed elite athletes and the type of exercise performed (swimming), which allowed as to analyze the applicability and effectiveness of tDCS in real-world sport context. Future studies should test if tDCS could be effective for the enhancement of endurance performance on longer duration exercises.

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Conclusions

Anodal tDCS applied over the primary motor cortex did not improve swimming performance in elite triathletes but resulted in a marked increase of vigor self-perception that was also maintained after exercise.

Acknowledgements

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Conflicts of interest

The authors declare no conflicts of interest.

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References

1. Gandevia SC. Spinal and supraspinal factors in human muscle fatigue. *Physiol Rev*. 2001;81(4):1725-1789.

2. Taylor JL, Amann M, Duchateau J, Meeusen R, Rice CL. Neural contributions to muscle fatigue: From the brain to the muscle and back again. *Med Sci Sports Exerc*. 2016;48(11):2294-2306. doi:10.1249/MSS.0000000000000923.

3. Davis NJ. Neurodoping: Brain stimulation as a performance-enhancing measure. *Sport Med*. 2013;43(8):649-653. doi:10.1007/s40279-013-0027-z.

4. Nitsche MA, Paulus W. Sustained excitability elevations induced by transcranial DC motor cortex stimulation in humans. *Neurology*. 2001;57(10):1899-1901. doi:10.1212/WNL.57.10.1899.

5. Vaseghi B, Zoghi M, Jaberzadeh S. Does anodal transcranial direct current stimulation modulate sensory perception and pain? A meta-analysis study. *Clin Neurophysiol*. 2014;125(9):1847-1858. doi:10.1016/j.clinph.2014.01.020.

6. Astokorki AHY, Mauger AR. Tolerance of exercise-induced pain at a fixed rating of perceived exertion predicts time trial cycling performance. *Scand J Med Sci Sport*. 2017;27(3):309-317. doi:10.1111/sms.12659.

7. Angius L, Hopker J, Mauger AR. The ergogenic effects of transcranial direct current stimulation on exercise performance. *Front Physiol*. 2017;8(FEB). doi:10.3389/fphys.2017.00090.

8. Vitor-Costa M, Okuno NM, Bortolotti H, et al. Improving cycling performance: Transcranial direct current stimulation increases time to exhaustion in cycling. *PLoS One*. 2015;10(12):1-15. doi:10.1371/journal.pone.0144916.

9. Okano AH, Fontes EB, Montenegro RA, et al. Brain stimulation modulates the autonomic nervous system, rating of perceived exertion and performance during maximal exercise. *Br J Sports Med*. 2015;49(18):1213-1218. doi:10.1136/bjsports-2012-091658.

10. Angius L, Mauger AR, Hopker J, Pascual-Leone A, Santarnecchi E, Marcora SM. Bilateral extracephalic transcranial direct current stimulation improves endurance performance in healthy individuals. *Brain Stimul*. 2018;11(1):108-117. doi:10.1016/j.brs.2017.09.017.

11. Angius L, Hopker JG, Marcora SM, Mauger AR. The effect of transcranial direct current stimulation of the motor cortex on exercise-induced pain. *Eur J Appl Physiol*. 2015;115(11):2311-2319. doi:10.1007/s00421-015-3212-y.

12. Barwood MJ, Butterworth J, Goodall S, et al. The Effects of Direct Current Stimulation on Exercise Performance, Pacing and Perception in Temperate and Hot Environments. *Brain Stimul*. 2016;9(6):842-849. doi:10.1016/j.brs.2016.07.006.

13. Russell M, Goodman T, Wang Q, Groshong B, Lyeth BG. Gender differences in current received during transcranial electrical stimulation. *Front Psychiatry*. 2014;5(AUG):1-7. doi:10.3389/fpsyt.2014.00104.

14. Batsikadze G, Moliadze V, Paulus W, Kuo MF, Nitsche MA. Partially non-linear stimulation intensity-dependent effects of direct current stimulation on motor cortex excitability in humans. *J Physiol*. 2013;591(7):1987-2000. doi:10.1113/jphysiol.2012.249730.

15. Gandiga PC, Hummel FC, Cohen LG. Transcranial DC stimulation (tDCS): A tool for double-blind sham-controlled clinical studies in brain stimulation. *Clin Neurophysiol*. 2006;117(4):845-850. doi:10.1016/j.clinph.2005.12.003.

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16. Brunoni AR, Amadera J, Berbel B, Volz MS, Rizzerio BG, Fregni F. A systematic review on reporting and assessment of adverse effects associated with transcranial direct current stimulation. *Int J Neuropsychopharmacol*. 2011;14(8):1133-1145. doi:10.1017/S1461145710001690.

17. Skorski S, Thompson KG, Keegan RJ, Meyer T, Abbiss CR. A monetary reward alters pacing but not performance in competitive cyclists. *Front Physiol*. 2017;8(SEP):1-11. doi:10.3389/fphys.2017.00741.

18. Terry PC, Lane AM, Fogarty GJ. Construct validity of the Profile of Mood States - Adolescents for use with adults. *Psychol Sport Exerc*. 2003;4(2):125-139. doi:10.1016/S1469-0292(01)00035-8.

19. Plews DJ, Laursen PB, Stanley J, Kilding AE, Buchheit M. Training adaptation and heart rate variability in elite endurance athletes: Opening the door to effective monitoring. *Sport Med*. 2013;43(9):773-781. doi:10.1007/s40279-013-0071-8.

20. Kovac S, Speckmann E, Gorji A. Uncensored EEG: the role of DC potentials in neurobiology of the brain. *Prog Neurobiol*. 2018. doi:10.1016/j.pneurobio.2018.02.001.

21. Heishman AD, Curtis MA, Saliba E, Hornett RJ, Malin SK, Weltman AL. Non-Invasive Assessment of Internal and External Player Load. *J Strength Cond Res*. 2018;(January):1. doi:10.1519/JSC.0000000000002413.

22. Hopkins WG. Estimating sample sizes for magnitude-based inference. *Sportscience*. 2006;10:63-70.

23. Hopkins W, Marshall SW, Batterham AM, Hanin J. Progressive statistics for studies in sports medicine and exercise science. *Med Sci Sports Exerc*. 2009;41(1):3-12. doi:10.1249/MSS.0b013e31818cb278.

24. Cohen J. *Statistical Power Analysis for the Behavioral Sciences*. 2nd ed. Hillsdale, New Jersey: Lawrence Erlbaum Associates; 1988. doi:10.1234/12345678.

25. Hopkins W. Spreadsheets for analysis of controlled trials, crossovers and time series. *Sportscience*. 2017;21:1-4.

26. Kim JH, Kim DW, Chang WH, Kim YH, Kim K, Im CH. Inconsistent outcomes of transcranial direct current stimulation may originate from anatomical differences among individuals: Electric field simulation using individual MRI data. *Neurosci Lett*. 2014;564:6-10. doi:10.1016/j.neulet.2014.01.054.

27. Angius L, Pageaux B, Hopker J, Marcora SM, Mauger AR. Transcranial direct current stimulation improves isometric time to exhaustion of the knee extensors. *Neuroscience*. 2016;339:363-375. doi:10.1016/j.neuroscience.2016.10.028.

28. Amann M, Venturelli M, Ives SJ, et al. Peripheral fatigue limits endurance exercise via a sensory feedback-mediated reduction in spinal motoneuronal output. *J Appl Physiol*. 2013;115(3):355-364. doi:10.1152/japplphysiol.00049.2013.

29. Thomas K, Goodall S, Stone M, Howatson G, Gibson ASC, Ansley L. Central and peripheral fatigue in male cyclists after 4-, 20-, and 40-km time trials. *Med Sci Sports Exerc*. 2015;47(3):537-546. doi:10.1249/MSS.0000000000000448.

30. Saw AE, Main LC, Gastin PB. Monitoring the athlete training response: Subjective self-reported measures trump commonly used objective measures: A systematic review. *Br J Sports Med*. 2016;50(5):281-291. doi:10.1136/bjsports-2015-094758.

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31. Zandi L, Rad LS. A comparison of the mood state profiles of winning and losing female athletes. *Eur J Exp Biol*. 2013;3(1):424-428.

32. Brandt R, Bevilacqua GG, Andrade A. Perceived Sleep Quality, Mood States, and Their Relationship with Performance among Brazilian Elite Athletes during a Competitive Period. *J Strength Cond Res*. 2017;31(4):1033-1039. doi:10.1519/JSC.0000000000001551.

33. Khedr EM, Omran EAH, Ismail NM, et al. Effects of transcranial direct current stimulation on pain, mood and serum endorphin level in the treatment of fibromyalgia: A double blinded, randomized clinical trial. *Brain Stimul*. 2017;10(5):893-901. doi:10.1016/j.brs.2017.06.006.

34. Kekic M, McClelland J, Bartholdy S, et al. Single-session transcranial direct current stimulation temporarily improves symptoms, mood, and self-regulatory control in bulimia nervosa: A randomised controlled trial. *PLoS One*. 2017;12(1):1-17. doi:10.1371/journal.pone.0167606.

35. Sehm B, Schafer A, Kipping J, et al. Dynamic modulation of intrinsic functional connectivity by transcranial direct current stimulation. *J Neurophysiol*. 2012;108(12):3253-3263. doi:10.1152/jn.00606.2012.

36. Lang N, Siebner HR, Ward NS, et al. How does transcranial DC stimulation of the primary motor cortex alter regional neuronal activity in the human brain? *Eur J Neurosci*. 2005;22(2):495-504. doi:10.1111/j.1460-9568.2005.04233.x.How.

37. Herrington JD, Mohanty A, Koven NS, et al. Emotion-modulated performance and activity in left dorsolateral prefrontal cortex. *Emotion*. 2005;5(2):200-207. doi:10.1037/1528-3542.5.2.200.

38. Okano AH, Machado DGS, Oliveira Neto L, et al. Can Transcranial Direct Current Stimulation Modulate Psychophysiological Response in Sedentary Men during Vigorous Aerobic Exercise? *Int J Sports Med*. 2017;38(7):493-500. doi:10.1055/s-0042-121897.

39. Montenegro RA, Farinatti P de TV, Fontes EB, et al. Transcranial direct current stimulation influences the cardiac autonomic nervous control. *Neurosci Lett*. 2011;497(1):32-36. doi:10.1016/j.neulet.2011.04.019.

40. Clancy JA, Johnson R, Raw R, Deuchars SA, Deuchars J. Anodal transcranial direct current stimulation (tDCS) over the motor cortex increases sympathetic nerve activity. *Brain Stimul*. 2014;7(1):97-104. doi:10.1016/j.brs.2013.08.005.

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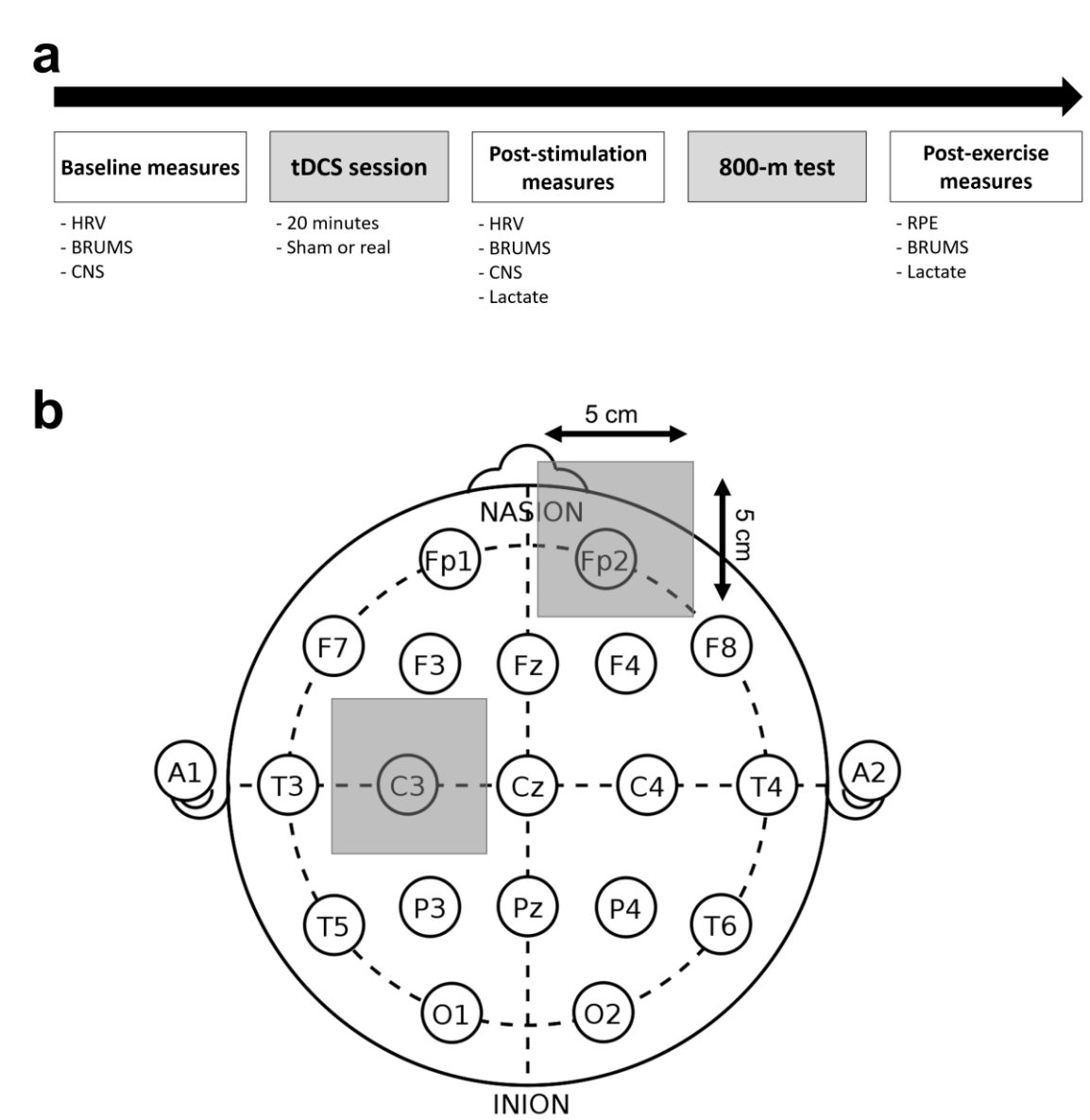

Figure 1. Schematic representation of the experimental protocol (**a**) and of the electrode montage (**b**). Panel (**b**) represents the placement of the electrodes on the international 10:20 system of electroencephalography. Abbreviations: BRUMS, Brunel Mood Scale; HRV, heart rate variability; RPE, rating of perceived exertion; tDCS, transcranial direct current stimulation.

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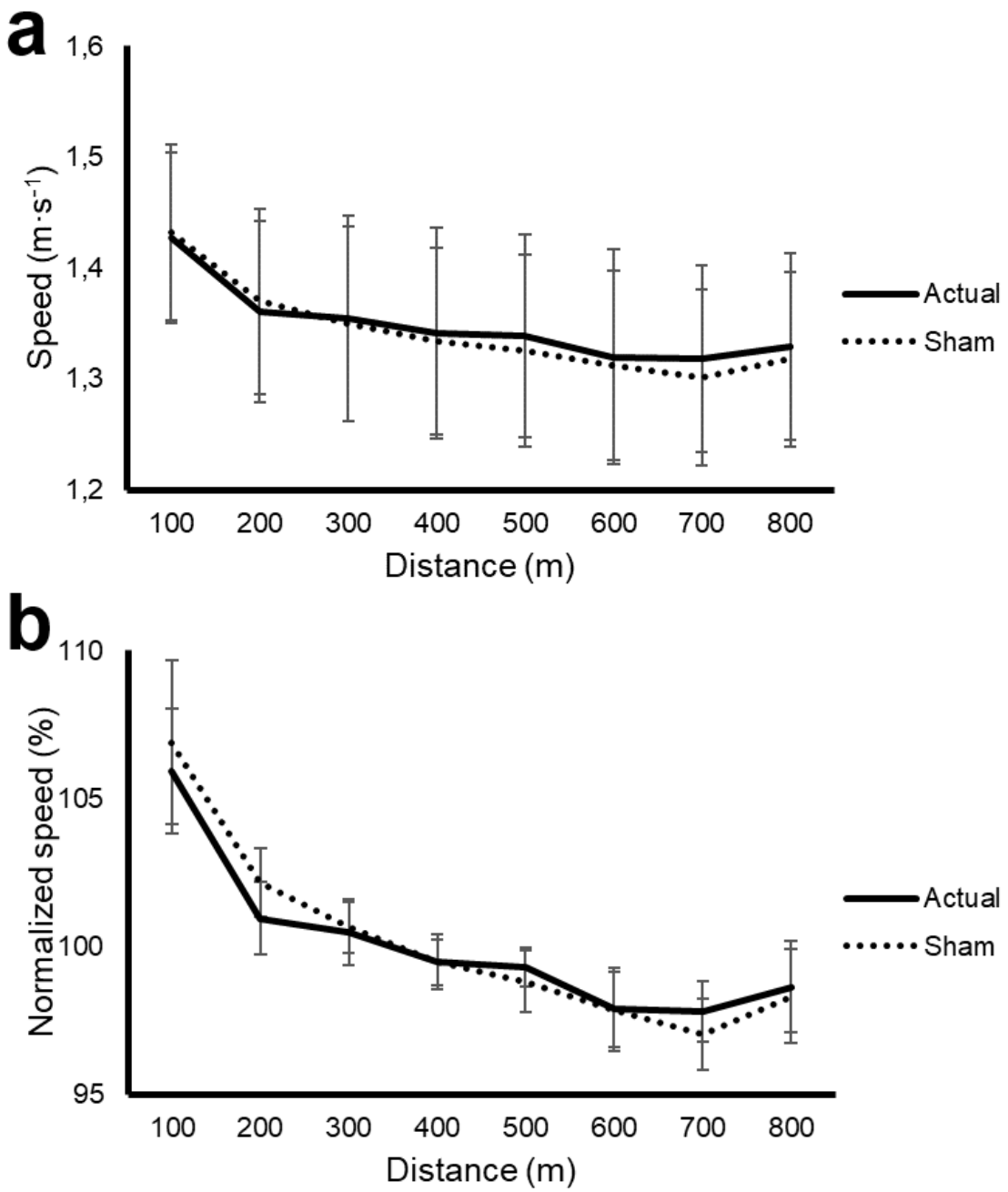

Figure 2. Raw (**a**) and normalized (**b**) speed during the 800-meter swimming test. Data are mean ± SD. There were no significant differences between conditions.

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Table 1. Side effects reported during the study during sham or actual transcranial direct current stimulation.

Adverse effect	n		Severity	
	Sham	Actual	Sham	Actual
Headache	1	0	0 (0–1)	-
Scalp pain	0	0	-	-
Scalp burn	0	0	-	-
Neck pain	0	0	-	-
Tingling	8	8	1 (1–3)	1 (1–3)
Skin redness	2	2	0 (0–1)	0 (0–1)
Sleepiness	1	1	0 (0–1)	0 (0–2)
Trouble concentrating	0	0	-	-
Mood change	0	0	-	-

Severity is presented as median (range). No significant differences (p>0.05) were observed in the prevalence of adverse effects (calculated with Fisher's exact test). Severity was self-rated as 0 (none), 1 (light), 2 (moderate) or 3 (severe).

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Table 2. Scores on the Brunel Mood Scale (BRUMS) scale

	Post-tDCS		**Post-exercise**		**Time Effect** (*p*-value)	**Condition x time Effect** (*p*-value)
	Sham	Actual	Sham	Actual		
Anger	-1 ± 1	0 ± 0	0 ± 2	1 ± 3	0.272	0.361
Confusion	0 ± 1	0 ± 2	0 ± 3	0 ± 2	0.716	0.694
Depression	1 ± 2	-1 ± 1	1 ± 2	1 ± 3	0.397	0.262
Fatigue	-1 ± 2	-1 ± 1	3 ± 3	3 ± 3	**0.001**	0.986
Tension	-1 ± 2	1 ± 1	-1 ± 2	-1 ± 2	0.114	0.405
Vigor	0 ± 1	2 ± 2*	-4 ± 3	-1 ± 4*	**0.003**	**0.012**

Data are mean ± SD change from baseline scores. Abbreviations: TDCs, transcranial direct current stimulation. Significant *p*-values are in bold. Significant differences between conditions determined with Bonferroni *post hoc* analysis: *$p<0.05$.

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